

Problem 19

Evaluate the integral:

$$\int \cot^2 x \, dx.$$

Solution

We aim to solve the integral:

$$\int \cot^2 x \, dx.$$

Step 1: Use the trigonometric identity for $\cot^2 x$

The trigonometric identity for $\cot^2 x$ is:

$$\cot^2 x = \csc^2 x - 1.$$

Substituting this identity into the integral, we have:

$$\int \cot^2 x \, dx = \int (\csc^2 x - 1) \, dx.$$

Step 2: Split the integral

The integral is now split into two simpler integrals:

$$\int \cot^2 x \, dx = \int \csc^2 x \, dx - \int 1 \, dx.$$

1. **First term: $\int \csc^2 x \, dx$:** The integral of $\csc^2 x$ is a standard result:

$$\int \csc^2 x \, dx = -\cot x.$$

2. **Second term: $\int 1 \, dx$:** The integral of 1 is:

$$\int 1 \, dx = x.$$

Step 3: Combine the results

Combining the results of the two integrals:

$$\int \cot^2 x \, dx = -\cot x - x + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$\int \cot^2 x \, dx = -\cot x - x + C.$$